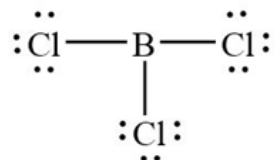


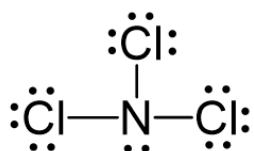
Exam 2 Review worksheet

1. For the following molecules, determine if the molecule is 1) nonpolar covalent, polar covalent, or ionic, and 2) list out all the intermolecular forces in them.

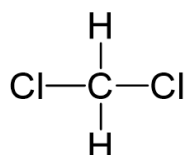
a.



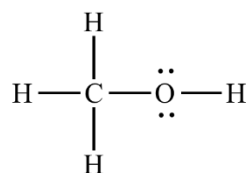
b.



c.



d.



2. In Q1, which molecule has ...

- Strongest intermolecular forces:
- Highest boiling point:
- Highest Vapor pressure:
- Highest viscosity:
- Highest Escape Kinetic energy:
- Which molecule will be miscible in oil (nonpolar solvent)?

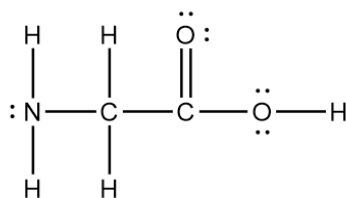
3. Describe the relationship between the following physical properties and intermolecular forces and explain it.

a. Boiling point

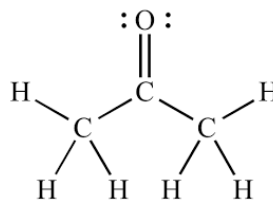
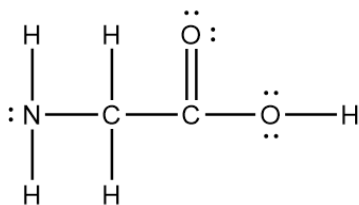
b. Vapor pressure

c. Escape kinetic energy

4. Circle out all H atoms that can form hydrogen bond in the following molecule:



5. Use the dashed line to draw one possible hydrogen bond between the following two molecule:



6. A sample of benzene is cooled from 320.0 °C to 205.0 °C and releases 29.4 kJ of heat. Draw the cooling curve for the process. How much benzene, in grams, is in this sample? The following physical data of benzene may be useful.

molar mass = 78.11 g/mol

$\Delta H_{\text{vap}} = 33.9 \text{ kJ/mol}$

$\Delta H_{\text{fus}} = 9.8 \text{ kJ/mol}$

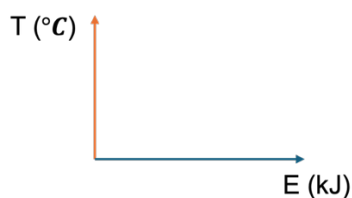
$C_{\text{liquid}} = 1.73 \text{ J/g}^\circ\text{C}$

$C_{\text{gas}} = 1.06 \text{ J/g}^\circ\text{C}$

$C_{\text{solid}} = 1.51 \text{ J/g}^\circ\text{C}$

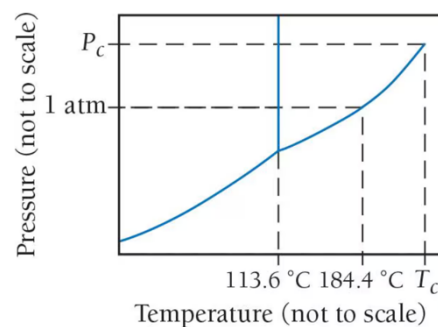
$T_{\text{melting}} = 279.0 \text{ }^\circ\text{C}$

$T_{\text{boiling}} = 353.0 \text{ }^\circ\text{C}$



7. Consider the phase diagram for I_2 below and answer the questions below:

- what is the normal boiling point for I_2 ?
- what state is present at room temperature and normal atmosphere pressure?
- how does the phase change from 0 atm to 1.5 atm at 100 °C?
- how does the phase change from 100 °C to 200 °C at 1 atm?

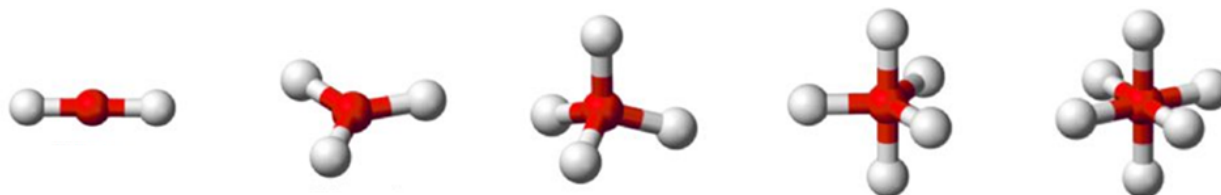


8. An aqueous Na_2SO_4 ($M=142.04 \text{ g/mol}$) solution has a molarity concentration of 1.81 M .
- Determine the molality concentration, m , of the solution. The density of the solution is 1.151 g/mL .
 - What is the boiling point, in $^\circ\text{C}$, of the solution? The boiling point of water is $100.^\circ\text{C}$ and $K_{b, \text{water}} = 0.512^\circ\text{C/m}$.
 - Explain why the boiling point is changed in Na_2SO_4 solution compared to pure water.
 - A methanol (CH_3OH) solution has the same molality concentration with the Na_2SO_4 solution. Which solution has a higher boiling point and why?
 - What is the vapor pressure, in torr, of the solution? $P^\circ_{\text{vap, H}_2\text{O}} = 31.82 \text{ torr}$.
 - Explain why the vapor pressure is changed in Na_2SO_4 solution compared to pure water.

$$\Delta T_f = iK_f m \quad T_{f, \text{soln}} = T_{f, \text{solvent}} - \Delta T_f \quad \Delta T_b = iK_b m \quad T_{b, \text{soln}} = T_{b, \text{solvent}} + \Delta T_b$$

Concentration Unit	Abbreviation	Concentration Formula
Molarity	M	$M = \frac{\text{moles of solute}}{\text{Liters of solution}}$
molality	m	$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$
Mole Fraction	X	$X_A = \frac{\text{moles of } A}{\text{total \# moles in solution}} = \frac{n_A}{n_A + n_B + n_C + \dots}$

Basic Geometric Shapes



List of Possible Molecular Geometry Names (Alphabetically listed)

Bent square	linear	octahedral	seesaw	square planar
pyramid	T-shaped	tetrahedral	trigonal bipyramidal	
trigonal planar	trigonal pyramidal			

Pauling Electronegativity Values for the Elements in the Periodic Table

H 2.20																He ---	
Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne ---
Na 0.93	Mg 1.31											Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar ---
K 0.82	Ca 1.0	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.0
Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.60	Mo 2.16	Tc 1.90	Ru 2.20	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.6
Cs 0.79	Ba 0.89	La-Lu 1.1-1.2	Hf 1.30	Ta 1.50	W 2.36	Re 1.90	Os 2.20	Ir 2.20	Pt 2.28	Au 2.54	Hg 2.00	Tl 1.62	Pb 1.87	Bi 2.02	Po 2.0	At 2.2	Rn ---
Fr 0.7	Ra 0.9	Ac-No 1.1-1.7															

Answer: Q2) d,d,a,d,d,a; Q6) 95.4 g; Q8a) 2.02m; Q8b) 103 °C; Q8e) 29.52 torr